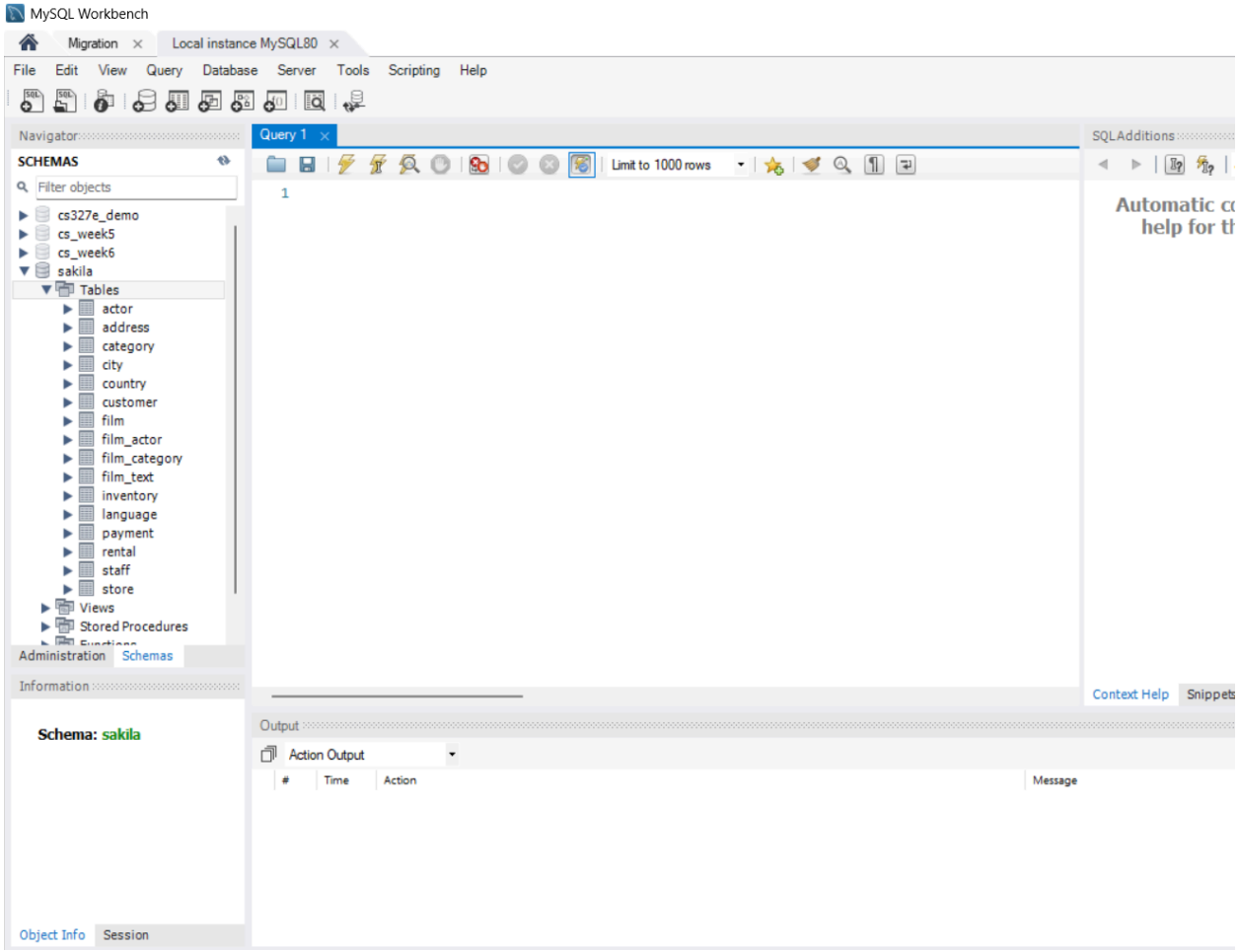
Madison Pascual
Kaleb Luna

Project 2 Results

Workbench Set-up



Describe the structure of tables in the Sakila DB

```
3 •  DESCRIBE actor;
4 •  DESCRIBE film_actor;
5 •  DESCRIBE payment;
6 •  DESCRIBE staff;
7 •  DESCRIBE rental;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
actor_id	smallint unsigned	NO	PRI	NULL	auto_increment
first_name	varchar(45)	NO		NULL	
last_name	varchar(45)	NO	MUL	NULL	
last_update	timestamp	NO		CURRENT_TIMESTAMP	DEFAULT_GENERATED on update CURRENT_TI...

```
3 •  DESCRIBE actor;
4 •  DESCRIBE film_actor;
5 •  DESCRIBE payment;
6 •  DESCRIBE staff;
7 •  DESCRIBE rental;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
actor_id	smallint unsigned	NO	PRI	NULL	
film_id	smallint unsigned	NO	PRI	NULL	
last_update	timestamp	NO		CURRENT_TIMESTAMP	DEFAULT_GENERATED on update CURRENT_TI...

Result Grid | Form Editor

```
3 •  DESCRIBE actor;
4 •  DESCRIBE film_actor;
5 •  DESCRIBE payment;
6 •  DESCRIBE staff;
7 •  DESCRIBE rental;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
payment_id	smallint unsigned	NO	PRI	NULL	auto_increment
customer_id	smallint unsigned	NO	MUL	NULL	
staff_id	tinyint unsigned	NO	MUL	NULL	
rental_id	int	YES	MUL	NULL	
amount	decimal(5,2)	NO		NULL	
payment_date	datetime	NO		NULL	
last_update	timestamp	YES		CURRENT_TIMESTAMP	DEFAULT_GENERATED on update CURRENT_TI.

Result Grid | Form Editor

```
3 •  DESCRIBE actor;
4 •  DESCRIBE film_actor;
5 •  DESCRIBE payment;
6 •  DESCRIBE staff;
7 •  DESCRIBE rental;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
staff_id	tinyint unsigned	NO	PRI	NULL	auto_increment
first_name	varchar(45)	NO		NULL	
last_name	varchar(45)	NO		NULL	
address_id	smallint unsigned	NO	MUL	NULL	
picture	blob	YES		NULL	
email	varchar(50)	YES		NULL	
store_id	tinyint unsigned	NO	MUL	NULL	
active	tinyint(1)	NO		1	
username	varchar(16)	NO		NULL	
password	varchar(40)	YES		NULL	
last_update	timestamp	NO		CURR...	DEFAULT_GEN...

Result Grid | Form Editor | Field Types

```
3 •  DESCRIBE actor;
4 •  DESCRIBE film_actor;
5 •  DESCRIBE payment;
6 •  DESCRIBE staff;
7 •  DESCRIBE rental;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
rental_id	int	NO	PRI	NULL	auto_increment
rental_date	datetime	NO	MUL	NULL	
inventory_id	mediumint unsigned	NO	MUL	NULL	
customer_id	smallint unsigned	NO	MUL	NULL	
return_date	datetime	YES		NULL	
staff_id	tinyint unsigned	NO	MUL	NULL	
last_update	timestamp	NO		CURRENT_TIMESTAMP	DEFAULT_GENERATED on update CURRENT_TI

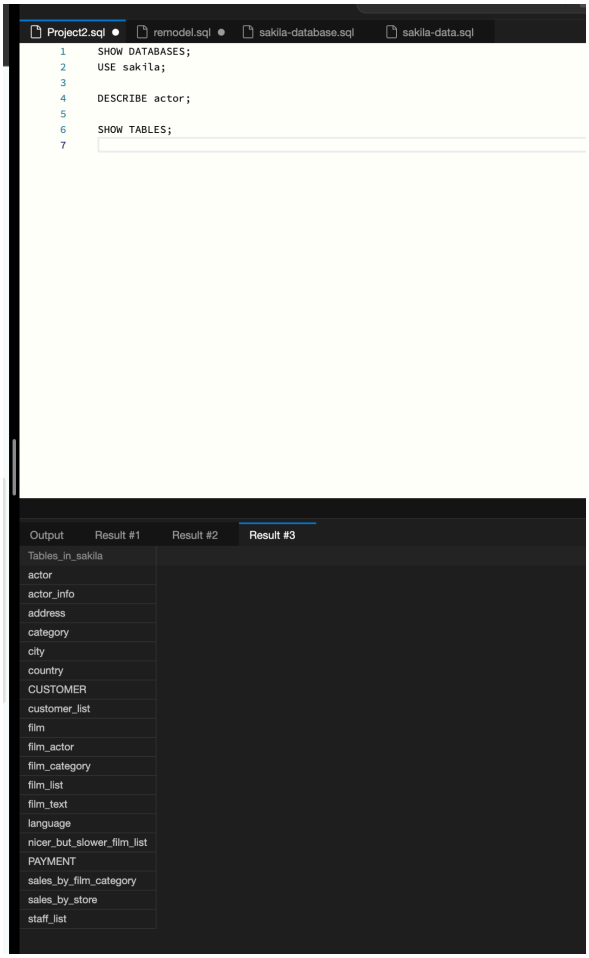
Result Grid | Form Editor

Modify the Sakila schema

```
12  --
15  ALTER TABLE STAFF DROP FOREIGN KEY fk_staff_store;
16  ALTER TABLE CUSTOMER DROP FOREIGN KEY fk_customer_store;
17
18  ALTER TABLE STORE DROP FOREIGN KEY fk_store_staff;
19
20  DROP TABLE STORE;
21
22
23  --
24  -- Dropping table 'staff'
25  -- First modify its child tables:
26  --   Table 'payment'
27  --   Table 'rental'
28  --   Table 'store' (done above, before dropping it)
29  --
30
31  ALTER TABLE PAYMENT DROP FOREIGN KEY fk_payment_staff;
32  ALTER TABLE RENTAL DROP FOREIGN KEY fk_rental_staff;
33  --  ALTER TABLE STORE DROP FOREIGN KEY fk_store_staff;
34
35  DROP TABLE STAFF;
36
37
38  --
39  -- Dropping table 'inventory'
40  -- First modify its child table:
41  --   Table 'rental'
42  --
43
44  -- Add DDL statements here ******
45  ALTER TABLE RENTAL DROP FOREIGN KEY fk_rental_inventory;
46  DROP TABLE INVENTORY;
47
48  --
49  -- Dropping table 'rental'
50  -- First modify its child table:
51  --   Table 'payment'
52  --
53
54  ALTER TABLE PAYMENT DROP FOREIGN KEY fk_payment_rental;
55  DROP TABLE RENTAL;
56
57  -- End
```

Output

```
#1: OK, 0 records retrieved in 73.351ms
```


Project2.sql
remodel.sql
sakila-database.sql
sakila-data.sql
1 SHOW DATABASES;
2 USE sakila;
3
4 DESCRIBE actor;
5
6 SHOW TABLES;
7
Output
Result #1
Result #2
Result #3
Tables_in_sakila
actor
actor_info
address
category
city
country
CUSTOMER
customer_list
film
film_actor
film_category
film_list
film_text
language
nicer_but_slower_film_list
PAYMENT
sales_by_film_category
sales_by_store
staff_list

Write 3 queries that use an INNER JOIN between two or more tables. The queries must also include a WHERE clause and ORDER BY clause. Feel free to include a LIMIT clause if the output returns > 10 rows. Add a short comment above each SQL statement

```sql
-- Display customers and their addresses in Texas ordered by last name
SELECT a.address_id, a.phone, c.customer_id, c.last_name
FROM ADDRESS a
INNER JOIN CUSTOMER c
    ON a.address_id = c.address_id
WHERE a.district = 'Texas'
ORDER BY c.last_name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

address_id	phone	customer_id	last_name
122	909029256431	118	CRUZ
10	860452626434	6	DAVIS
405	775235029633	400	HARDISON
310	262088367001	305	MCCRARY
567	239357986667	561	STILL

```sql
-- Display the actors who appeared in film 7, ordered by last name
SELECT a.actor_id, a.first_name, a.last_name, f.film_id
FROM ACTOR a
INNER JOIN FILM_ACTOR f
    ON a.actor_id = f.actor_id
WHERE f.film_id = 7
ORDER BY a.last_name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

actor_id	first_name	last_name	film_id
185	MICHAEL	BOLGER	7
170	MENA	HOPPER	7
162	OPRAH	KILMER	7
99	JIM	MOSTEL	7
133	RICHARD	PENN	7

```sql
-- Display cities located in the United States, ordered alphabetically
SELECT ci.city_id, ci.city, co.country_id, co.country
FROM CITY ci
INNER JOIN COUNTRY co
    ON ci.country_id = co.country_id
WHERE co.country = 'United States'
ORDER BY ci.city
LIMIT 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

city_id	city	country_id	country
11	Akron	103	United States
33	Arlington	103	United States
41	Augusta-Richmond County	103	United States
42	Aurora	103	United States
65	Bellevue	103	United States

Write 3 queries that use an OUTER JOIN between two or more tables. The queries must also include a WHERE clause and ORDER BY clause. Feel free to include a LIMIT clause if the output returns > 10 rows. Add a short comment above each SQL statement.

```sql
-- Display all actors whos last names start with P and any films they have
SELECT a.actor_id, a.first_name, a.last_name, fa.film_id, f.title
FROM ACTOR a
LEFT JOIN FILM_ACTOR fa
    ON a.actor_id = fa.actor_id
LEFT JOIN FILM f
    ON fa.film_id = f.film_id
WHERE a.last_name LIKE 'P%'
ORDER BY a.last_name
LIMIT 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

actor_id	first_name	last_name	film_id	title
21	KIRSTEN	PALTROW	6	AGENT TRUMAN
21	KIRSTEN	PALTROW	87	BOONDOCK BALLROOM
21	KIRSTEN	PALTROW	88	BORN SPINAL
21	KIRSTEN	PALTROW	142	CHICKEN HELLFIGHTERS
21	KIRSTEN	PALTROW	159	CLOSER BANG
21	KIRSTEN	PALTROW	179	CONQUERER NUTS
21	KIRSTEN	PALTROW	253	DRIFTER COMMANDMENTS
21	KIRSTEN	PALTROW	281	ENCINO ELF
21	KIRSTEN	PALTROW	321	FLASH WARS

Result 25

```sql
-- Display customers with payments greater than 11, ordered by last name
SELECT c.customer_id, c.first_name, c.last_name, p.payment_id, p.amount
FROM CUSTOMER c
RIGHT JOIN PAYMENT p
    ON c.customer_id = p.customer_id
WHERE p.amount > 11
ORDER BY c.last_name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_id	first_name	last_name	payment_id	amount
591	KENT	ARSENAULT	15821	11.99
196	ALMA	AUSTIN	5281	11.99
362	NICHOLAS	BARFIELD	9803	11.99
116	VICTORIA	GIBSON	3146	11.99
237	TANYA	GILBERT	6409	11.99
13	KAREN	JACKSON	342	11.99
305	RICHARD	MCCRARY	8272	11.99
592	TERRANCE	ROUSH	15850	11.99
204	ROSEMARY	SCHMIDT	5550	11.99
195	VANESSA	SIMS	5280	11.99

```sql
-- Display films and their associated languages, ordered by language
SELECT f.film_id, f.title, f.original_language_id, l.language_id, l.name
FROM FILM f
LEFT JOIN LANGUAGE l
    ON f.language_id = l.language_id
WHERE f.film_id > 0
ORDER BY l.name
LIMIT 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

film_id	title	original_language_id	language_id	name
1	ACADEMY DINOSAUR	NULL	1	English
2	ACE GOLDFINGER	NULL	1	English
3	ADAPTATION HOLES	NULL	1	English
4	AFFAIR PREJUDICE	NULL	1	English
5	AFRICAN EGG	NULL	1	English
6	AGENT TRUMAN	NULL	1	English
7	AIRPLANE SIERRA	NULL	1	English
8	AIRPORT POLLOCK	NULL	1	English
9	ALABAMA DEVIL	NULL	1	English
10	ALADDIN CALENDAR	NULL	1	English

Write 2 queries that use a SELF JOIN on any table. Feel free to include a LIMIT clause if the output returns > 10 rows. Add a short comment above each SQL statement.

```sql
-- Display pairs of addresses located in the same city
SELECT a1.address_id AS address_1, a2.address_id AS address_2, a1.city_id
FROM ADDRESS a1
JOIN ADDRESS a2
    ON a1.city_id = a2.city_id
WHERE a1.address_id < a2.address_id
ORDER BY a1.city_id
LIMIT 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

address_1	address_2	city_id
335	543	42
1	3	300
256	517	312
2	4	576

Result 71

```sql
-- Display movies that have the exact same runtime
SELECT f1.length, f1.title as movie_1, f2.title as movie_2
FROM FILM f1
JOIN FILM f2
    ON f1.length = f2.length
WHERE f1.film_id < f2.film_id
ORDER BY f1.length
LIMIT 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

length	movie_1	movie_2
46	ALIEN CENTER	IRON MOON
46	ALIEN CENTER	KWAI HOMEWARD
46	ALIEN CENTER	LABYRINTH LEAGUE
46	ALIEN CENTER	RIDGEMONT SUBMARINE
46	IRON MOON	KWAI HOMEWARD

Result 69